

Signal, Noise, and Detection in CCD Photometry

A first-principles account of where the noise in a CCD measurement comes from, how it sets the signal-to-noise ratio, and what that implies for how you should actually plan an exposure.

A CCD does not hand you a flux. It hands you a number of electrons collected in a set of pixels over some time, and that number carries an uncertainty. Everything that matters observationally — whether an object is detected, how precisely its magnitude is known, whether it is smarter to take one long exposure or many short ones — follows from tracking that uncertainty honestly from its physical origin down to the final signal-to-noise ratio, SNR. That is the whole content of these notes: no empirical fudge factors, just Poisson statistics applied to four physically distinct sources of electrons, added the way independent random variables are supposed to be added.

Notation used throughout

One deliberate choice, kept consistent for the rest of the document: every noise term is called σ (a standard deviation), never N — because N is also the natural symbol for a photon count, and reusing it for both “count” and “noise” is a standing source of confusion. Rates are all called R with a descriptive subscript, and the exposure is split into N_{exp} repeats of length t_{exp} rather than the instrument-jargon NDIT/DIT.

Symbol	Meaning	Units
R_{src}	electron rate from the source, integrated over the photometric aperture	$\text{e}^- \text{s}^{-1}$
R_{sky}	sky-background electron rate, per pixel	$\text{e}^- \text{s}^{-1} \text{pix}^{-1}$
R_{dark}	dark-current electron rate, per pixel	$\text{e}^- \text{s}^{-1} \text{pix}^{-1}$
n_{pix}	number of pixels in the photometric aperture ($\propto \text{seeing}^2$)	pixels
$t_{\text{exp}}, N_{\text{exp}}$	duration of one exposure; number of exposures	s ; —
$t = N_{\text{exp}} t_{\text{exp}}$	total exposure time	s
$S = R_{\text{src}} t$	total signal: source electrons collected	e^-
σ_{RON}	read noise, per pixel per readout	e^-
σ_x	standard deviation (“noise”) contributed by process x	e^-
$\text{SNR} = S/\sigma_{\text{tot}}$	signal-to-noise ratio	—
$m, \delta m$	magnitude; uncertainty on the magnitude	mag

Setting up the measurement

A photon hitting the detector is converted to a photoelectron with some probability η , the quantum efficiency (typically $\eta \sim 0.5\text{--}0.85$ for a CCD). We will not carry η around explicitly: since it multiplies every photon rate by the same constant factor, we agree from the outset to count everything in *detected electrons*, and let η be absorbed into the rates. This is a bookkeeping choice, not a physical assumption — but it means every rate below ($R_{\text{src}}, R_{\text{sky}}, R_{\text{dark}}$) is already “post-QE.”

Over a total exposure time t , the source deposits a total signal

$$S = R_{\text{src}} t. \quad (1)$$

In practice t is built out of N_{exp} individual exposures of length t_{exp} each:

$$t = N_{\text{exp}} t_{\text{exp}}, \quad S = R_{\text{src}} N_{\text{exp}} t_{\text{exp}}. \quad (2)$$

This distinction — total time t versus how it is *partitioned* into N_{exp} chunks — turns out to be the whole story later, because some noise terms care only about t and one cares specifically about N_{exp} .

What noise buys you: from SNR to an error bar on the magnitude

The noise σ is the 1σ uncertainty on S : a measurement is reported as $S \pm \sigma$. The signal-to-noise ratio is defined as

$$\text{SNR} = \frac{S}{\sigma_{\text{tot}}}, \quad (3)$$

so $1/\text{SNR}$ is just the fractional uncertainty on the measurement. It is the single number that tells you how much you can trust what you measured:

SNR	fractional error $1/\text{SNR}$	what it means
2–3	33–50%	object barely detected
5	20%	detected — you can believe you're seeing something real
10	10%	a real measurement
100	1%	excellent

Because astronomers report brightness in magnitudes rather than flux, it is worth converting once and for all. By definition $m = -2.5 \log_{10}(S)$ (up to zero-point constants that don't depend on S and therefore drop out of any error propagation). If the true count deviates from S by the noise σ , the induced shift in magnitude is

$$\delta m = 2.5 \log_{10} \left(\frac{S + \sigma}{S} \right) = 2.5 \log_{10} \left(1 + \frac{\sigma}{S} \right). \quad (4)$$

For $\sigma/S \ll 1$ this linearizes ($\ln(1+x) \approx x$) to the familiar rule of thumb

$$\delta m \approx \frac{2.5}{\ln 10} \cdot \frac{\sigma}{S} = \frac{1.0857}{\text{SNR}}. \quad (5)$$

CHECKED NUMERICALLY

The exact expression (Eq. 4) and the linear approximation (Eq. 5) agree well at high SNR but diverge once the fractional error becomes large — exactly where a linearization should be expected to fail. The comparison in Fig. 1 confirms the rule of thumb $\delta m \sim 1/\text{SNR}$ is safe once SNR is above about 10, but at SNR = 2–3 (barely detected) the exact logarithmic form should be used instead.

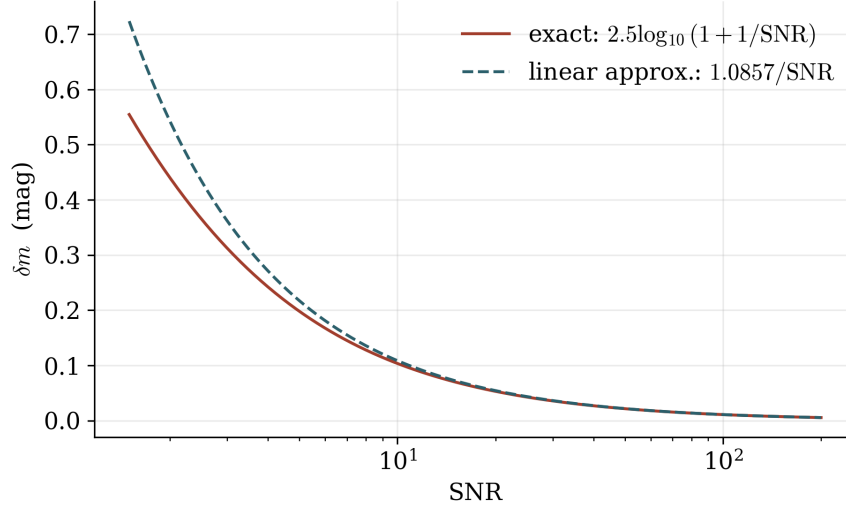


Figure 1: Exact magnitude error versus the linear small-error approximation, as a function of SNR. The two curves are indistinguishable above $\text{SNR} \sim 20$ and diverge below $\text{SNR} \sim 5$.

These numbers are not an arbitrary convention, and it's worth pausing to justify them rather than just citing the table. The table actually answers two different questions, and its two halves answer different ones: rows 2–3 and 5 answer “is this real, or could it just be noise?”; rows 10 and 100 answer “given that it's real, how precisely do I know its brightness?” They happen to share the same axis (SNR) because both questions are governed by the same ratio, but the reasoning behind each is worth making explicit.

Rows 2–3 and 5: SNR as a detection significance

Assumption. The total noise σ_{tot} is well approximated by a Gaussian (normal) distribution. This is a genuinely new assumption, not a repeat of the Poisson assumption from the next section — but it is a well justified one: σ_{tot} is built from a sum of many independent contributions (electrons from many photons, many pixels, many reads), and the Central Limit Theorem says that a sum of many independent random contributions tends toward a Gaussian regardless of the shape of the individual distributions, provided no single term dominates the sum. (A single Poisson variable itself becomes Gaussian at large μ , since its skewness scales as $1/\sqrt{\mu}$ and vanishes as μ grows.) This is precisely what licenses treating “SNR” as a literal number of standard deviations, i.e. speaking of a “ $k\sigma$ detection.”

With that assumption in hand, pose the question precisely: suppose that, in truth, there is no object at this location — the real signal is $S = 0$, and every electron measured is background and noise. The measured count still fluctuates, by construction, with standard deviation σ_{tot} around zero. The probability that noise alone produces an apparent excess of at least $k\sigma_{\text{tot}}$ is the upper tail of the Gaussian,

$$P(k) = \int_k^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 1 - \Phi(k), \quad (6)$$

where Φ is the standard normal CDF. Because $\text{SNR} = k$ in this hypothetical no-signal case, Eq. 6 converts an observed SNR directly into a *false-alarm probability*: the chance that what you're look-

ing at is pure noise dressed up to look like a detection.

CHECKED NUMERICALLY

Evaluating Eq. 6 numerically: $P(2) = 2.3 \times 10^{-2}$ (1 in 44), $P(3) = 1.4 \times 10^{-3}$ (1 in 741), $P(5) = 2.9 \times 10^{-7}$ (1 in 3.5 million), $P(10) = 7.6 \times 10^{-24}$. The probability falls by roughly an order of magnitude or more with each extra unit of k — Fig. 2 shows the full curve.

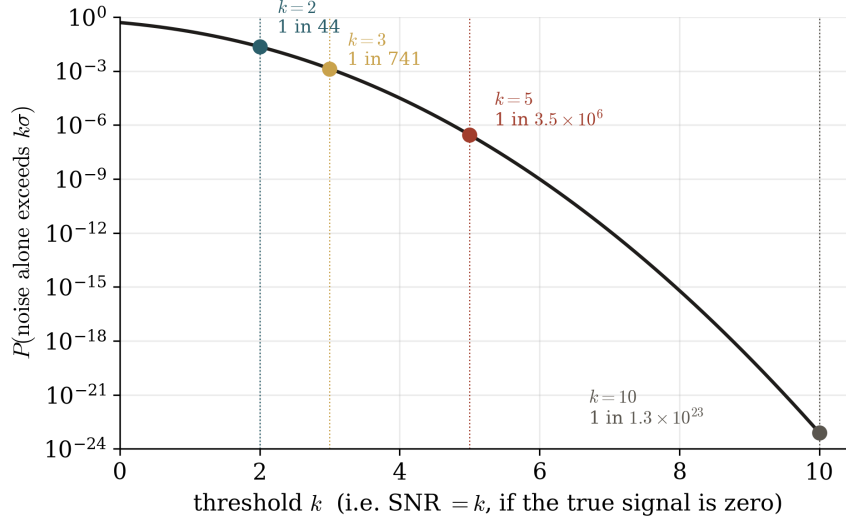


Figure 2: The probability that noise alone (Gaussian, mean zero) produces an apparent excess of at least $k\sigma$, plotted on a log scale, for the case of a single pre-specified measurement. This is exactly why $\text{SNR} = 2\text{--}3$ is only “barely” trusted (there is still a percent-level chance it’s a fluctuation) while $\text{SNR} = 5$ is treated as essentially certain: a one-in-3.5-million fluke is not a serious competing explanation.

This is enough to explain rows 2–3 and 5 for a single, pre-chosen measurement — checking the brightness of one specific, already-known star, say. But an important qualification is needed before this reasoning is complete.

THE CATCH: FALSE ALARMS MULTIPLY WITH THE NUMBER OF TRIES

Eq. 6 is a *per-trial* probability. If instead of checking one pre-chosen location you scan an entire image of n_{trial} independent pixels or apertures looking for whatever turns up (a blind search for new sources), the chance that *at least one* of them exceeds the threshold by chance alone is, for small $P(k)$, approximately

$$P_{\text{any}}(k) \approx n_{\text{trial}} P(k) \quad (7)$$

(a union bound: the probability of at least one rare event among many independent tries is approximately the sum of the individual probabilities). For a 4000×4000 image, $n_{\text{trial}} = 1.6 \times 10^7$, and at $k = 3$ the expected number of pure-noise “detections” is $n_{\text{trial}} P(3) \approx 2.2 \times 10^4$ — utterly swamped by false positives, useless as a blind detection threshold. At $k = 5$ the

expectation drops to ~ 4.6 ; by $k \approx 5.5$ – 6 it is comfortably below 1. This is exactly why blind searches (transient surveys, particle-physics discovery claims) conventionally demand “ 5σ ” rather than “ 3σ ”: the higher bar is not tradition for its own sake, it is what keeps the expected count of noise-only false positives, summed over every one of the many independent trials in the search, at or below about one. A targeted measurement of a single known object is not a blind search — it is one trial, not n_{trial} — which is why the same table casually lists $\text{SNR} = 2$ – 3 as “barely detected” rather than “worthless”: it is worthless *as a blind-search claim*, but usable as a single, pre-registered check.

Rows 10 and 100: SNR as a measurement precision

By $\text{SNR} = 10$ the false-alarm probability from Eq. 6 is already astronomically small ($\sim 10^{-24}$ at $k = 10$, from the box above) — so rows 10 and 100 are not answering “is it real?” at all; that question is already settled. They answer a completely different one: given that the object is real, how well do you know *how bright* it is? That question is answered directly by the fractional error $1/\text{SNR}$ derived above, and by its magnitude-space version, Eq. 5: $\text{SNR} = 10$ means a 10% flux error, about 0.1 mag — usable for a rough photometric point. $\text{SNR} = 100$ means 1%, about 0.01 mag — the level needed for precision work such as transit photometry or millimag-level variability studies.

So the full table is really two different axes glued together by a shared symbol: the low end (2–3, 5) is a statement about statistical confidence that a source exists at all (governed by Gaussian tail probabilities, and secretly dependent on how many independent trials you searched over), while the high end (10, 100) is a statement about how precisely you know its brightness once existence is no longer in question (governed directly by $1/\text{SNR}$, already derived above).

Where the noise actually comes from

σ is not one thing. It is the quadrature sum of several statistically independent contributions, and the physics of each is different — which is exactly why the answer to “should I take one long exposure or ten short ones?” depends on which term dominates.

1. Shot noise: why counting photons is noisy even for a perfectly steady source

Before writing down $\sigma = \sqrt{\mu}$, it’s worth asking why there should be any noise at all in the first place. If a star’s true brightness is perfectly constant, why isn’t the electron count perfectly reproducible from one exposure to the next? The answer is that light is not a continuous fluid; it arrives at the detector as a stream of discrete quanta (photons), and *even a perfectly stable source emits those quanta at random, independent times*. This randomness is not a flaw in the instrument — it is a property of the light itself, and no amount of engineering removes it. That irreducible floor is what “shot noise” means (the name comes from the analogous effect in an electrical current, which is carried by discrete electrons arriving at random times rather than by a smooth flow of charge). Because it is fundamental rather than instrumental, it is the right place to start: everything else in this section (sky, dark, read noise) is a variation on the same counting argument, or, in the case of read noise, a genuinely different mechanism introduced for contrast.

Assumption. Photon arrivals from the source are independent of one another, and occur at a

constant mean rate λ (photons per second) over the exposure. This is an excellent approximation for essentially all classical (thermal or incoherent) astronomical sources; it is exactly the statement that the light is *not* correlated in a way that makes photons arrive in bunches or avoid each other.

From independent arrivals to a distribution. Split the total exposure time t into N equal subintervals of width $\Delta t = t/N$, with N large enough that Δt is small. In each subinterval, the probability of detecting exactly one photon is $p = \lambda \Delta t$, and the probability of detecting more than one vanishes as $\Delta t \rightarrow 0$; so each subinterval is a Bernoulli trial, and — because the arrivals are independent by assumption — the total count n over all N subintervals follows a Binomial distribution,

$$P(n) = \binom{N}{n} p^n (1-p)^{N-n}, \quad p = \lambda \Delta t = \frac{\mu}{N}, \quad (8)$$

where $\mu \equiv \lambda t$ is the expected total count over the full exposure. Now take the limit $N \rightarrow \infty$ (finer and finer subintervals) at fixed $\mu = Np$. Using $N!/(N-n)! \rightarrow N^n$ and $(1 - \mu/N)^{N-n} \rightarrow e^{-\mu}$ as $N \rightarrow \infty$, the binomial collapses to the *Poisson distribution*:

$$P(n; \mu) = \frac{\mu^n e^{-\mu}}{n!}. \quad (9)$$

This is the only distribution consistent with independent, constant-rate arrivals — it is not an extra assumption, it is a consequence of the one assumption already made above.

Mean and variance, derived, not asserted. The mean of Eq. 9 is

$$\langle n \rangle = \sum_{n=0}^{\infty} n \frac{\mu^n e^{-\mu}}{n!} = \mu e^{-\mu} \sum_{n=1}^{\infty} \frac{\mu^{n-1}}{(n-1)!} = \mu e^{-\mu} e^{\mu} = \mu, \quad (10)$$

where the sum was reindexed with $k = n - 1$ and recognized as the exponential series. The same trick applied to $n(n-1)$ gives the second factorial moment,

$$\langle n(n-1) \rangle = \sum_{n=0}^{\infty} n(n-1) \frac{\mu^n e^{-\mu}}{n!} = \mu^2 e^{-\mu} \sum_{n=2}^{\infty} \frac{\mu^{n-2}}{(n-2)!} = \mu^2, \quad (11)$$

so that $\langle n^2 \rangle = \langle n(n-1) \rangle + \langle n \rangle = \mu^2 + \mu$, and the variance is

$$\text{Var}(n) = \langle n^2 \rangle - \langle n \rangle^2 = \mu^2 + \mu - \mu^2 = \mu. \quad (12)$$

The mean and the variance of a Poisson variable are the same number, μ — a special, non-obvious property of this particular distribution, not a generic fact about randomness. The noise (standard deviation) follows immediately:

$$\sigma = \sqrt{\text{Var}(n)} = \sqrt{\mu}. \quad (13)$$

This single relation, applied three times to three different electron sources (source, sky, dark current, each its own independent Poisson process with its own μ), generates three of the four terms in the noise budget.

CHECKED NUMERICALLY

For $\mu = 10$, summing Eq. 9 directly (truncated at $n = 200$, far past where the terms become negligible) gives $\sum_n P(n) = 1.0000$, $\langle n \rangle = 10.000$, and $\text{Var}(n) = 10.000$ — confirming the closed-form results above rather than just asserting them.

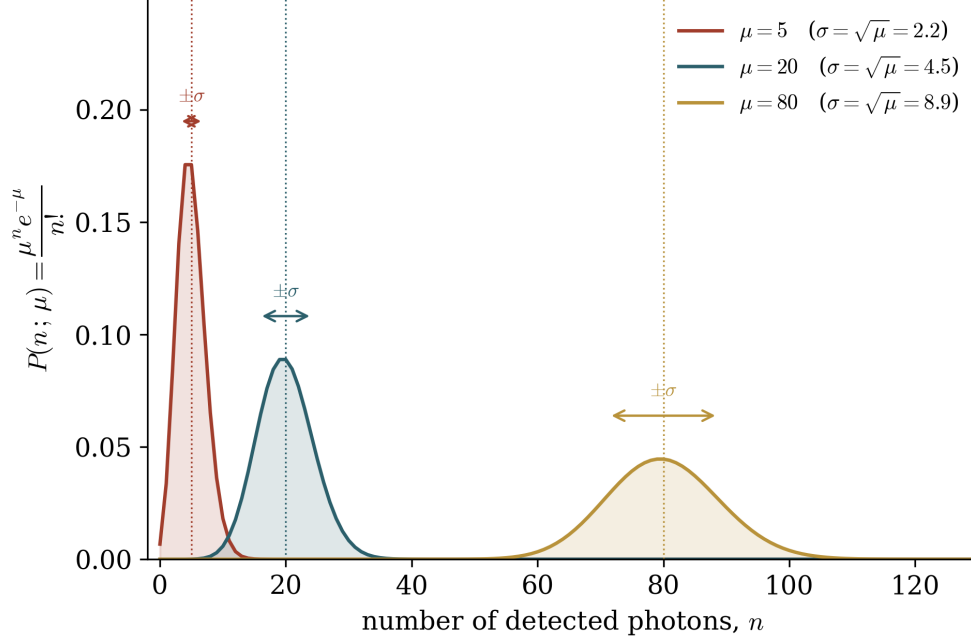


Figure 3: The Poisson distribution $P(n; \mu)$ for three mean photon counts. The horizontal arrows mark $\pm\sigma = \pm\sqrt{\mu}$ around each mean. Two things to read off directly: the *absolute* width grows with μ (a brighter source really does have a noisier raw count), while the width *relative to the mean*, $\sigma/\mu = 1/\sqrt{\mu}$, visibly shrinks — the three curves get narrower in proportion to their height as μ increases. This is precisely why $\text{SNR} = \mu/\sigma = \sqrt{\mu}$ improves as you collect more light, and why it does so only as a square root rather than linearly: the noise grows too, just more slowly than the signal.

Object shot noise. The object itself delivers $S = R_{\text{src}} t$ electrons, and by the Poisson rule above,

$$\sigma_{\text{src}} = \sqrt{S} = \sqrt{R_{\text{src}} t}. \quad (14)$$

Sky shot noise. You cannot measure the object without also collecting sky background in the same aperture, and that sky signal has to be estimated and subtracted — but its own shot noise cannot be subtracted away. With R_{sky} the sky rate per pixel and n_{pix} the number of pixels in the photometric aperture (set by the seeing: $n_{\text{pix}} \propto \text{seeing}^2$), the total sky electrons collected under the object are $n_{\text{pix}} R_{\text{sky}} t$, so

$$\sigma_{\text{sky}} = \sqrt{n_{\text{pix}} R_{\text{sky}} t}. \quad (15)$$

Dark-current shot noise. Thermal electrons generated inside the silicon itself, independent of any

light, accumulate at a rate R_{dark} per pixel, hence

$$\sigma_{\text{dark}} = \sqrt{n_{\text{pix}} R_{\text{dark}} t}. \quad (16)$$

FLAGGING AN AMBIGUITY IN THE UNITS

Dark current is often quoted per hour, purely because it's a small number over a typical exposure. But $\sigma_{\text{dark}} = \sqrt{n_{\text{pix}} R_{\text{dark}} t}$ is only dimensionally consistent if R_{dark} and t use the *same* time unit. If a data sheet gives R_{dark} in $\text{e}^- \text{hr}^{-1}$ and t is integrated in seconds, convert first: $R_{\text{dark}} [\text{e}^- \text{s}^{-1}] = R_{\text{dark}} [\text{e}^- \text{hr}^{-1}] / 3600$. Throughout these notes, R_{src} , R_{sky} , R_{dark} are all per second, matching t in seconds — a convention made explicit here rather than left implicit.

2. Read noise: not a shot-noise term

Every time the chip is read out, the on-chip amplifier adds a noise contribution σ_{RON} (electrons), fixed per pixel per readout, independent of exposure time or flux. It doesn't come from counting anything, so it doesn't obey the $\sqrt{\mu}$ scaling — it is simply a fixed number added once per read. Reading n_{pix} pixels once contributes noise $\sqrt{n_{\text{pix}}} \sigma_{\text{RON}}$ (independent per-pixel noises add in quadrature). Reading out N_{exp} times over the course of building up the total exposure means paying this penalty N_{exp} times:

$$\sigma_{\text{read}} = \sqrt{n_{\text{pix}} N_{\text{exp}}} \sigma_{\text{RON}}. \quad (17)$$

This is the crucial structural difference from the three shot-noise terms: $\sigma_{\text{read}}^2 \propto N_{\text{exp}}$, not t . Splitting a fixed total time into more sub-exposures costs additional read noise, for free, every time.

Assembling the total: the master SNR equation

FIRST PRINCIPLES DERIVATION: WHY VARIANCES ADD FOR INDEPENDENT NOISE SOURCES

Let X and Y be two random variables representing independent physical noise processes with expected values $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$ and variances $\sigma_X^2 = \text{Var}(X)$, $\sigma_Y^2 = \text{Var}(Y)$. The variance of their sum $Z = X + Y$ is defined by:

$$\text{Var}(X + Y) = \mathbb{E} \left[((X + Y) - \mathbb{E}[X + Y])^2 \right] = \mathbb{E} \left[((X - \mu_X) + (Y - \mu_Y))^2 \right]. \quad (18)$$

Expanding the algebraic square inside the expectation:

$$((X - \mu_X) + (Y - \mu_Y))^2 = (X - \mu_X)^2 + (Y - \mu_Y)^2 + 2(X - \mu_X)(Y - \mu_Y). \quad (19)$$

Applying the linearity of the mathematical expectation operator $\mathbb{E}[\cdot]$:

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E} [(X - \mu_X)^2] + \mathbb{E} [(Y - \mu_Y)^2] + 2 \mathbb{E} [(X - \mu_X)(Y - \mu_Y)] \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y). \end{aligned} \quad (20)$$

Because the physical mechanisms generating astronomical source photons, sky background photons, thermal dark current, and amplifier readout noise are statistically *independent*, their

joint probability density factorizes: $p(x, y) = p_X(x) p_Y(y)$. The covariance therefore vanishes identically:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)] \mathbb{E}[(Y - \mu_Y)] = 0 \cdot 0 = 0. \quad (21)$$

By mathematical induction, for N mutually independent noise contributions $X_1, X_2, \dots, X_N$:

$$\text{Var}\left(\sum_{i=1}^N X_i\right) = \sum_{i=1}^N \text{Var}(X_i) \implies \sigma_{\text{tot}}^2 = \sum_{i=1}^N \sigma_i^2 \iff \sigma_{\text{tot}} = \sqrt{\sum_{i=1}^N \sigma_i^2}. \quad (22)$$

This rigorous result proves why independent noise contributions must be added in quadrature, establishing the fundamental basis of the CCD noise budget.

Because the four contributions arise from independent physical processes, their variances add:

$$\sigma_{\text{tot}}^2 = \sigma_{\text{src}}^2 + \sigma_{\text{sky}}^2 + \sigma_{\text{dark}}^2 + \sigma_{\text{read}}^2. \quad (23)$$

Substituting the four expressions derived above,

$$\text{SNR} = \frac{R_{\text{src}} t}{\sqrt{R_{\text{src}} t + n_{\text{pix}} R_{\text{sky}} t + n_{\text{pix}} R_{\text{dark}} t + n_{\text{pix}} N_{\text{exp}} \sigma_{\text{RON}}^2}} \quad (24)$$

with $t = N_{\text{exp}} t_{\text{exp}}$. Every quantity here is a property of the target (R_{src}), the sky (R_{sky}), the instrument (n_{pix} , R_{dark} , σ_{RON}), or the observing strategy (t , N_{exp}). This is the formula an exposure-time calculator evaluates; everything below is what it looks like in the three limits where one term wins.

The three regimes, and why they dictate how you observe

Bright source (photon / shot-noise limited)

When the target is bright enough that $R_{\text{src}} t$ swamps sky, dark, and read noise, the master equation collapses to

$$\text{SNR} = \frac{R_{\text{src}} t}{\sqrt{R_{\text{src}} t}} = \sqrt{R_{\text{src}} t} \implies \text{SNR} \propto \sqrt{t}. \quad (25)$$

This is the regime of bright standard stars: the practical limit is not reaching adequate SNR but avoiding saturation of the detector.

Sky-background limited

For a faint object against a bright sky, $R_{\text{sky}} \gg R_{\text{src}}$, so the sky term dominates the sum under the square root:

$$\text{SNR} \approx \frac{R_{\text{src}} t}{\sqrt{n_{\text{pix}} R_{\text{sky}} t}} \implies \text{SNR} \propto \frac{\sqrt{t}}{\sqrt{n_{\text{pix}}}}. \quad (26)$$

Two consequences follow directly, and both are operationally important:

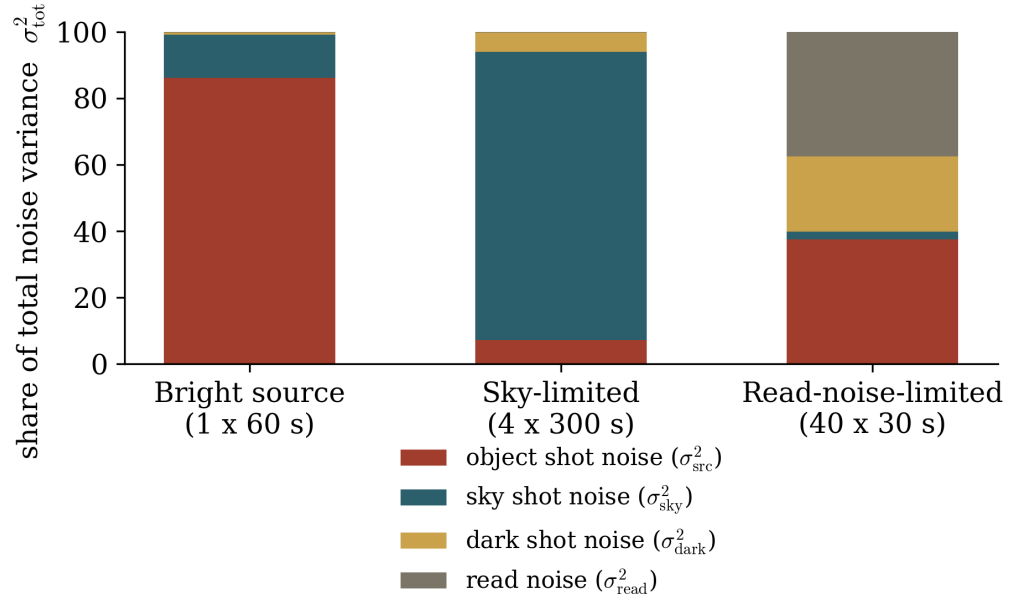


Figure 4: The four variance terms of the master equation, normalized to sum to 100%, evaluated for three representative setups (parameters chosen to be illustrative, not from a specific instrument). This is exactly what “which regime am I in?” means in practice: look at which slice dominates.

- Since $n_{\text{pix}} \propto \text{seeing}^2$, tighter seeing (smaller effective aperture) directly buys SNR — this is why image quality matters even when you’re not resolution-limited scientifically.
- SNR depends only on the *total* time $t = N_{\text{exp}} t_{\text{exp}}$, not on how it is split. One 3000 s exposure and ten 300 s exposures give (to first order) the same SNR, provided each sub-exposure is individually sky-dominated. In practice, splitting is usually the better choice anyway — it lets you reject cosmic rays and build a better flat field — and it costs essentially nothing.

This is the regime of broadband optical imaging and low-resolution optical spectroscopy, and — in a more extreme form — near-infrared imaging, where the sky is so bright that long individual exposures would saturate the array within seconds; there, splitting into many short exposures is not a choice but a necessity, and the physics above says it’s harmless to SNR.

Read-noise limited

When sky and dark are both small (narrow-band imaging, high-resolution or echelle spectroscopy, where the light per pixel is intrinsically low), the read-noise term wins:

$$\text{SNR} \approx \frac{R_{\text{src}} t}{\sqrt{n_{\text{pix}} N_{\text{exp}}} \sigma_{\text{RON}}} \implies \text{SNR} \propto \frac{1}{\sqrt{n_{\text{pix}}}} \cdot \frac{1}{\sqrt{N_{\text{exp}}}}. \quad (27)$$

Here the two factors give two distinct, actionable levers:

- **Binning.** n_{pix} is the number of pixels actually *read out*, not an immutable property of the source. Binning 2×2 on-chip before readout reduces n_{pix} by a factor of 4 and hence buys a factor $\sqrt{4} = 2$ in SNR — genuinely free signal-to-noise, at the cost of spatial (or spectral) resolution. Keep at least ~ 2 binned pixels across the seeing disk (imaging) or across a spectral line (spectroscopy); binning past that throws away resolution you needed.
- **Minimize N_{exp} .** Since $\text{SNR} \propto N_{\text{exp}}^{-1/2}$ at fixed total time, t_{exp} should be as long as possible (equivalently N_{exp} as small as possible) — the opposite strategy from the sky-limited case. The practical ceiling on t_{exp} is usually set by cosmic-ray accumulation (commonly ~ 45 min for CCDs), not by any noise term in this equation; if cosmic rays genuinely don’t matter to the science, there’s no formal reason not to go longer still.

This regime governs narrow-band imaging (sky reduced to a couple of electrons per pixel, exposures up to ~ 1 hr) and echelle spectroscopy, where light is dispersed over so many resolution elements that the sky contribution per pixel becomes negligible.

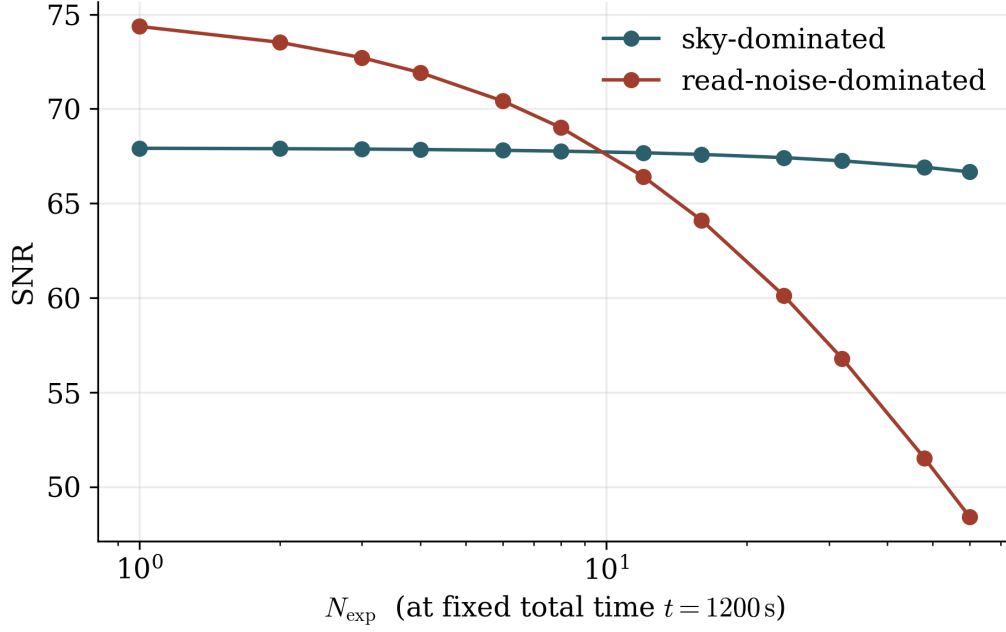


Figure 5: SNR versus N_{exp} at *fixed total exposure time* $t = N_{\text{exp}} t_{\text{exp}} = 1200 \text{ s}$, computed directly from the master equation (Eq. 24, no approximation) for two illustrative parameter sets. **Sky-dominated** ($R_{\text{src}} = 50$, $R_{\text{sky}} = 30 \text{ e}^- \text{ s}^{-1} \text{ pix}^{-1}$, $n_{\text{pix}} = 20$, $\sigma_{\text{RON}} = 5 \text{ e}^-$): the curve is essentially flat — splitting the exposure is nearly free. **Read-noise-dominated** ($R_{\text{src}} = 5$, $R_{\text{sky}} = 0.05$, $n_{\text{pix}} = 6$, $\sigma_{\text{RON}} = 5 \text{ e}^-$): SNR falls visibly as N_{exp} grows, because each additional readout adds a fixed noise penalty unrelated to how much light was actually collected. Same total integration time, same instrument-class read noise — opposite conclusions about how to schedule the exposure.

Summary: reading the regime off the noise budget

All three limits are the same equation; they differ only in which of four physically distinct terms — object shot noise, sky shot noise, dark shot noise, read noise — happens to dominate the sum in the denominator. That, and nothing more exotic, is why “should I take one exposure or many?” and “should I bin?” have different answers in different kinds of observations: the questions are really asking which term in σ_{tot}^2 you’re fighting.

Regime	Dominant term	SNR scaling	Strategy
Bright source	σ_{src}^2	$\sqrt{t}$	watch for saturation, not SNR
Sky-limited	σ_{sky}^2	$\sqrt{t}/\sqrt{n_{\text{pix}}}$	splitting exposures is free; good seeing helps directly
Read-noise-limited	σ_{read}^2	$n_{\text{pix}}^{-1/2} N_{\text{exp}}^{-1/2}$	bin pixels; keep t_{exp} as long as cosmic rays allow

Physical content and worked numbers adapted from O. Hainaut, “Signal, Noise and Detection” (ESO, 2005); the notation, the units flag on dark current, and the figures above are this document’s own.